sligocki About

Lucy's Moonlight: The 5% Champion

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There is a 5% chance that we have a new BB(6) champion! It's a machine discovered by Racheline which she's named "Lucy's Moonlight". It is a probviously halting tetrational Cryptid. In other words, it's a machine that we believe will eventually halt (with 100% chance via a probabilistic approximation), yet proving it requires either solving a Collatz-like problem or simulating it for a "tetrational" number of steps. This is very similar to the Mother of Giants with the exception that if the Mother of Giants TMs (quasi-)halt, then they are known to beat the current BBB(5) champion, whereas if Lucy's Moonlight halts there is only a 5% chance that it beats the current 10↑↑15 BB(6) champion, in the other 95% cases it will halt in fewer steps.

Machine

The machine is `1RB0RD_0RC1RE_1RD0LA_1LE1LC_1RF0LD_1RZ0RA`:

	0	1
A	1RB	0RD
B	0RC	1RE
C	1RD	0LA
D	1LE	1LC
E	1RF	0LD
F	1RZ	0RA

It was discovered by Racheline on 1 Mar 2025 on the bbchallenge Discord (Discord link) and has bbchallenge wiki page: Lucy's Moonlight.

Analysis

Let

$$C(a, b) = 0^\infty \;\; 1011^a \;\; 1^b \;\; 10 \quad \text{C>} \quad 0^\infty$$

then, it can be proven that

$$\begin{array}{lll} \text{Start} & \xrightarrow{2} & C(0, 0) \\ \\ C(a+1, \quad 3k) & \to & C(a, 8k+6) \\ C(a+2, \quad 3k+1) & \to & C(a, 8k+16) \\ C(a+2, \quad 3k+2) & \to & C(a, 8k+22) \\ \\ C(0, \qquad 3k+1) & \to & C(0, 8k+5) \\ C(0, \qquad 3k+2) & \to & C(0, 8k+5) \\ \\ C(0, \qquad 3k) & \to & C(2k, 8) \\ C(1, \qquad 3k+2) & \to & C(2k+4, 8) \\ \\ C(1, \qquad 3k+1) & \to & \text{Halt}(6k+14) \end{array}$$

Where, as usual, these rules apply for all non-negative integer variable assignments ($a, k \geq 0$) and Halt(n) means that the TM halts with n non-zero symbols on the tape.

I've split these rules up into blocks to emphasize the different phases or types of rules:

Main Phase:

$$\begin{array}{lll} C(a+1, \quad 3k) & \to & C(a, 8k+6) \\ C(a+2, \quad 3k+1) & \to & C(a, 8k+16) \\ C(a+2, \quad 3k+2) & \to & C(a, 8k+22) \end{array}$$

This repeatedly applies a Collatz-like iteration, reducing a at each step until a is too small.

Reset Rules:

$$\begin{array}{lll} C(0, \quad 3k) & \to & C(2k, 8) \\ C(1, \quad 3k+2) & \to & C(2k+4, 8) \end{array}$$

If a, b match either of these two patterns then the TM resets back to a new main phase.

Zero Phase:

$$
\begin{aligned}
C(0, \quad 3k+1) & \quad \to \quad C(0, 8k+5) \\
C(0, \quad 3k+2) & \quad \to \quad C(0, 8k+5)
\end{aligned}
$$

If $a = 0$ and $3 \nmid b$ then the TM iterates a different Collatz-like function until it reaches a config where $3 \mid b$ and then resets.

Halt Rule:

$$C(1, \quad 3k+1) \quad \to \quad \text{Halt}(6k+14)$$

Finally, if it hits this specific configuration, the TM halts.

Trajectory

Note that after each reset, we end up in a config $C(a, 8)$. Let's call each of these configs "checkpoints". Following each checkpoint, the Lucy's Moonlight will either Halt, Reset or (theoretically) get stuck forever iterating Zero Phase.

Starting from a blank tape, the TM enters config $C(0, 0)$ after 2 steps and then follows the trajectory:

$$
\begin{array}{llllll}
C(0,0) & \to & C(0,8) & \to & C(0,21) & \\
 & \to & C(14,8) & \to & \cdots & \to \quad C(1,16934) \\
 & \to & C(11292,8) & \to & \cdots & \to \quad C(0, \approx 10^{2902.09}) \\
 & \to & C(\approx 10^{2901.92}, 8) & \to & \cdots &
\end{array}
$$

So, the first four checkpoints occur with a values:

$$
\begin{aligned}
c_0 & = & 0 \\
c_1 & = & 14 \\
c_2 & = & 11\,292 \\
c_3 & \approx & 10^{2901.92}
\end{aligned}
$$

▶ Exact value for c3

At this point it would require simulating more than 10^{2901} iterations of Main Phase in order to compute what happens next.

Probabilistic Argument

As with previous Collatz-like TMs, we will now consider a probabilistic model where we basically assume that the Collatz iterations operate like a (pseudo-)random number generator which has an equal chance of choosing any remainder modulo 3 for b at each step.

So,

- When $a \geq 2$, there will be a 1/3 chance of reducing a by 1, and 2/3 chance of reducing a by 2.
- When $a = 1$, there is a 1/3 chance of each: Halting immediately, resetting, entering Zero phase.
- When $a = 0$, Zero Phase will always eventually end, leading to a reset.

In this model, after every checkpoint, there are only two options: reset or halt. Probability of halt is 1/3 times the probability that we reach the $a = 1$ case (i.e. that it wasn't skipped by a Main Phase jump). Let $P(n)$ be the probability that starting from some config $C(a, b)$ (for any $a > n$) it reaches $C(a - n, b')$ (via Main Phase rules alone). Well, the only way for this iteration to not reach $C(a - n, b')$ is if it skipped over this a value. In other words:

$$P(n) = 1 - \frac{2}{3} P(n - 1)$$

This recurrence has its only fixed point at $\frac{3}{5}$ and if $P(n) = \frac{3}{5} + \epsilon$ then $P(n + 1) = \frac{3}{5} - \frac{2}{3}\epsilon$ so the probabilities approach this fixed point exponentially quickly: $\left|P(n) - \frac{3}{5}\right| = \frac{2}{5}\left(\frac{2}{3}\right)^n$. This is already tiny by $n = 20$ ($\left|P(14) - \frac{3}{5}\right| < 0.0002$), by the time you get to $n \approx 10^{2901.92}$ the discrepancy is minute.

Therefore, starting at any large checkpoint, there is a $\frac{1}{5}$ chance of halting this iteration ($\frac{3}{5}$ chance of reaching $a = 1$ and $\frac{1}{3}$ chance to halt if reaching $a = 1$) and a $\frac{4}{5}$ chance of resetting. And thus, the chance that it resets at least r times starting from a given checkpoint is $\left(\frac{4}{5}\right)^r$

Champion Chance

How many resets do we need to beat the current champion?

First note that both the reset rules produce a checkpoint config with even first parameter. So, without loss of generality, we can consider two sequential checkpoints as:

$$C(2a_n, 8) \to C(2a_{n+1}, 8)$$

In order to reset we know that this included **at least** a_n iterations of Main Phase and so

$$a_{n+1} \geq \frac{8}{3}\left(\frac{8}{3}\right)^{a_n}$$

If $C(2a_n, 8) \to Halt(x)$ then there is similar growth:

$$x \geq 16 \left(\frac{8}{3}\right)^{a_n}$$

If we let $b_k = \frac{3}{8} a_k$ then

$$b_{n+1} \geq \left(\frac{8}{3}\right)^{\frac{8}{3} b_n} > 10^{b_n}$$

Starting from $C(c_3, 8)$ with $c_3 > 8 \times 10^{2901}$ we get $b_3 = \frac{3}{16} c_3 > 10^{2901} > 10 \uparrow\uparrow 2.5$ (using fractional up-arrow notation). If this is followed by r resets, then we will reach

$$b_{r+3} > 10 \uparrow\uparrow (r + 2.5)$$

and when it halts there will be $> 10 \uparrow\uparrow (r + 3.5)$ ones on the tape. The current champion writes $< 10 \uparrow\uparrow 15.60466$ ones, so if $r \geq 13$, Lucy's Moonlight will beat it.[1]

Therefore, in this probabilistic model, knowing that it reaches checkpoint $C(> 8 \times 10^{2901}, 8)$, the chance that Lucy's Moonlight will write more ones than than Pavel's current champion is:

$$\left(\frac{4}{5}\right)^{13} \approx 5.50\%$$

If we could somehow accelerate the simulation and demonstrate that it reset at least once more, then that probability would increase slightly to:

$$\left(\frac{4}{5}\right)^{12} \approx 6.87\%$$

Footnotes

1. It appears that $r = 12$ resets followed by a halt will not be quite enough to snatch the championship, since it would have $\approx 10 \uparrow\uparrow 15.54$ ones, just shy of the $10 \uparrow\uparrow 15.60466$ needed for victory. But you'd have to be a little more careful with the analysis to ensure this is true. ↩

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sligocki

Shawn Ligocki
sligocki@gmail.com

sligocki